



Combining official and Google Trends data to forecast the Italian youth unemployment rate

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ABSTRACT

The increased availability of online information in recent years has aroused interest in the possibility of deriving indications for many kinds of phenomena. In the more specific economic and statistical context, numerous studies suggest the use of online search data to improve the forecasting and nowcasting of official economic indicators with a view to increasing the promptness of their circulation. The purpose of this work is to investigate if the use of big data can improve the forecasting of the youth unemployment rate – estimated in Italy on a monthly basis by the Italian National Institute of Statistics – by means of time series models. The time series used are those of the Google Trends query share for the keyword *offerte di lavoro* (job offers) and the official labour force survey data for the Italian youth unemployment rate since 2004. Two different models are estimated: an ARIMA model using only the official youth unemployment rate series and a VAR model combining the former series with the Google Trends query share. The results show that the use of Google Trends information leads to an average decrease in the forecast error.

1. Introduction

The paper addresses a problem that is closely connected with the forecasts of official statistical indicators produced monthly by national departments of statistics in all the European countries based on data gathered through sample surveys. For all these indicators, including the youth unemployment rate, it is useful to have forecasts that give some idea of the variations to be expected in the short term, not only because they can serve as preliminary estimates of the phenomenon examined but also because they can be incorporated into econometric models to provide indications of the prospects for the economic system as a whole. The Italian National Institute of Statistics (ISTAT) produces such forecasts by means of statistical models that make no use of information from external sources and rely only on the data provided by its own sample surveys.

The purpose of the experiment presented in this paper is to ascertain if the combined use of information from Google Trends (GT) and the ISTAT labour force survey makes any improvement to the forecasting of the youth unemployment rate in terms of the accuracy and timeliness of the estimates.

Many scholars and researchers have become aware in recent years that the vast amount of information to be derived from the mass of online search data available could prove useful in the study of social

phenomena (Askatas and Zimmermann, 2015; Chamberlin, 2010; Choi and Varian, 2012; Daas et al., 2015; Einav and Levin, 2014; Guzman, 2011; Marchetti et al., 2015). This is because the information that people disclose about their needs using the Internet (Ettredge et al., 2005) can shed light on the variability of numerous phenomena under examination. A review of the application of these types of data in different fields can be found in Hassani and Silva (2015).

With particular reference to Google Trends data, it is easy to state that the relevance and potential of this tool lies in their characteristic of containing a huge amount of information that is easily accessible yet difficult to obtain in another way. This represents a great opportunity especially for statistical purposes, and the interest of the scientific community is evident by the amount of work proposed in literature over the past 10 years. Only a few of the most recent studies are involved in small area estimation problems (Porter et al., 2013; Rao and Molina, 2015, p. 159–170) in the spread of infectious diseases (Anggraeni and Aristiani, 2017; Teng et al., 2017), in numerous economic contexts such as consumer purchases (Schmidt and Vosen, 2013), housing market (Limnios and You, 2016), tourist inflows (Dinis et al., 2017), forecasting energy (Hassani and Silva, 2016), credit developments (Burdeau and Knitzler, 2017), and social phenomena (Nghiem et al., 2016; Nixon, 2016; Parker et al., 2017).

In the official statistics field, some studies have been carried out to

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assess if online search data – in particular Google Trends data – can be used to facilitate the estimation of the phenomena of interest for the national statistical institutes or to produce additional information (Charles-Edwards, 2016; D'Alò et al., 2015; Falorsi et al., 2017; Kristoufek et al., 2016).

Regarding the phenomenon studied in this work, numerous authors suggest the use of Internet data to forecast unemployment (Anvik and Gjelstad, 2010; Askitas and Zimmermann, 2009; D'Amuri and Marcucci, 2017; Falorsi et al., 2015; McLaren and Shanbhogue, 2011), and the results show that they can indeed be regarded as useful in the estimation procedure.

The monthly estimates of Italian employment and unemployment published by the ISTAT have been accompanied regularly in the mass media by a particular focus on youth unemployment, the levels of which rose steadily during the recent economic crisis and are markedly above the European average, hence the interest in providing more timely and reliable information regarding this aspect.

Moreover, the official estimate of the youth unemployment rate in Italy is based on a monthly sample survey and generally becomes available about 30 days after the end of the month in question due to the time required to gather and process the relevant data. It could prove to be useful in many situations to have reliable indications of the results of an ongoing survey before all of its phases have been completed. In other words, the need may arise for a preview or nowcast (Choi and Varian, 2012) of the youth unemployment rate for which the survey is being carried out in the same period. The problem therefore is to produce estimates that are available while the actual survey is still under way – or in any case in a short space of time – to provide reliable information on the indicator in the current month and its variation with respect to the previous one (Ayoubkhani, 2012; D'Alò et al., 2006).

It should be noted that for Italy, as well as the other European countries, the information required by the European Community for purposes of economic analysis is vast and drawn from a variety of large-scale surveys based both on samples and on censuses. These requirements also are laid down in the European Community regulations on short-term business statistics in force since August 2005. The same date saw the launch of the Action Plan on Economic and Monetary Union (EMU) Statistical Requirements by Eurostat and the Central European Bank with the involvement of the European national departments of statistics in an effort to reduce the time required for the production and circulation of the most important indicators that are essential to the short-term analysis of the European economy.

The purpose of this work is to ascertain if the use of information from Google Trends together with data from the time series of the ISTAT labour force survey can improve the accuracy and timeliness of the youth unemployment rate forecast (Barreira et al., 2013; Fondeur and Karamè, 2013) and make it possible to provide an “immediate” forecast for use, even if only as a pointer, while awaiting the official figures.

Google Trends information is subjected to no form of quality control and, in our opinion, can be used solely as a sort of “snapshot”, providing indications about the phenomenon of interest and its evolution in time and space. As C.F. Citro (2014) observes:

Statistical agencies need, above all, sources of data that cover a known population with error properties that are reasonably well understood and that are not likely to change under their feet – characteristics that are not inherent in such data sources as autonomous interactions with websites on the Internet. There are, however, at least two ways in which household survey-based statistical agency programs could obtain an “edge” from non-traditional sources: one is to improve timeliness for preliminary estimates of key statistics; and the other is to provide leading indicators of social change (e.g., the emergence of new occupations and fields of training) that alert statistical agencies to needed changes in their concepts and measures.

Goel et al. (2010) provide a useful survey of the work undertaken in this area and describe some of the limitations of web search data. As they point out, such data are readily obtainable and often helpful in making forecasts but may not provide dramatic increases in predictability.

As the use of the Internet for job seeking is more widespread among the younger population, the trend in the monthly number of job searches made through Google can provide useful information for forecasting the youth unemployment rate. The basic idea is that an increase in this rate is accompanied by an increase in the number of young people seeking employment opportunities through online searches. As shown in Section 2.1, this idea is borne out by the results of a specific query in ISTAT's monthly survey on the labour force.

Google Trends data corresponding to the keyword “*offerte di lavoro*” (job offers) are used in this paper as auxiliary information to forecast the Italian youth unemployment rate. They belong to the sphere of what is known in the literature as “big data” (Ceron et al., 2014; Choi and Varian, 2012; Einav and Levin, 2014), which have the characteristic of being immediately available and capable of supplying an up-to-date picture of social and economic phenomena. As Chamberlin (2010) writes:

As this search data from Google Trends is available in real time, any significant relationship could be potentially exploited for nowcasting. Google Trends data may also be used informally to pick up on changing patterns and rising trends in search queries and the implications they have for types of economic activity and spending.

Moreover, Google Trends data generally can be used at a very low cost, which constitutes a further reason for the interest in the sphere of official statistical surveys.

However, there are some aspects in the use of GT data processing that cannot be left out. Among these, one of the most important is the choice of keywords to be used for the selection of data sets. Different keywords lead to different results; however, if the selection criterion derives from an adequate knowledge of the phenomenon under study and if you carry out controls on a large number of keywords and their combinations, you can obtain useful results.

In extracting information from large amounts of data, the errors that can be made are multiple (Braaksma and Zeelemborg, 2015; Nuti et al., 2014; Zeelemborg and Braaksma, 2017, p. 274–296) and many are the consequences of an inappropriate choice of keywords. The most probable error, especially in the event that GT information is used to make forecasts by means of regression models – as is the case in this paper – is to run into spurious correlations. The GT series generally is extracted on the basis of the correlation that it shows with the phenomenon under study. If two or more phenomena are statistically correlated, however, it does not necessarily mean that there is a direct cause-effect link between them because this correlation can be completely random (i.e. spurious) or dependent on a further variable in common.

To reduce this risk, different keywords or combinations of keywords are chosen, the variability of the series extracted from Google Trends is studied as keywords inserted in the query shares change, and finally for the extracted series the statistical significance of the correlation between these and the variable being studied is verified.

To evaluate the utility of GT data in forecasting the Italian youth unemployment rate, a comparison was developed between results obtained by means of an ARIMA model (Box et al., 2007), making no use of auxiliary GT information, and a vector autoregressive (VAR) model (Lutkepohl, 2007), harnessing the correlation of official and Internet search data.

The ARIMA methodology is the one already employed by the ISTAT to obtain forecasts in the sphere of national accounts and indicators of trends in the Italian economy. These forecasts are always short term (no longer than 3 months) and provide information on the expected tendency of the economic indicator in a given month of the year with respect to its value as published based on survey data 2 or 3 months earlier. In other words, the purpose of the forecasts is to provide

advance information on possible monthly and quarterly variations.

In general, choosing between different models means choosing which of the different representations of reality is the best; to do this, it is necessary to adopt a criterion on the basis of which to evaluate the models. In this work, the criterion uses several indicators: the significance of the estimated parameters, the goodness of fit of the model, and different measurements of the forecasting error.

The paper is organized as follows. Section 2 describes Italian youth unemployment by means of labour force survey (LFS) information, presents the time series used to forecast the youth unemployment rate, and briefly describes the forecasting models. Section 3 presents the results of the model estimation, and Section 4 puts forward some conclusions.

2. Material and methods

The phenomenon of youth unemployment in Italy, its scale, and the connections with Internet searches for job opportunities are briefly outlined in Section 2.1. The time series used for the estimation of the forecasting models is the monthly ISTAT series on the youth unemployment rate (YUR) in Italy and the monthly Google Trends (GT) series for the keyword “*offerte di lavoro*” (job offers). The latter refers to the number of searches made per month on Google specifying the words *offerte di lavoro*. Both series are described in detail in Section 2.2.

It should be noted that the use of this keyword is justified on the grounds of its popularity among competing job search-related keywords, as can be ascertained by comparing its relative incidence with that of other search terms (D’Amuri and Marcucci, 2017). Furthermore, its broad definition makes it ideally robust to irrelevant strong variations connected to the demand or supply for specific subgroups of job seekers. Although the fact that not all workers have access to the Internet also must be taken into account, this is a minor issue in view of its ever-increasing use as a way of seeking employment, especially by the young. Moreover, the two monthly series of the LFS regarding young people – the overall sample and the subsample of online job seekers – show marked similarities.

It should be noted that many other sources of the big data type are available to provide information on unemployment, including the number of positions posted by employers on their own websites or with Internet-based companies that mediate between employers and employees. Although these provide useful information with respect to categories of workers or geographical areas (Larsen et al., 2015), GT presents many characteristics that make it preferable to other sources in relation to the problem addressed here.

First and foremost, GT supplies information in the form of time series, which is obviously necessary to produce forecasts by means of the statistical models employed. Moreover, it is a source of homogeneous information that is available on one platform and provides information produced based on only one criterion. In other words, there is no need for procedures to merge information from different online “containers”, which generally is a very expensive and time-consuming process that also involves the risk of aggregating the data incorrectly. Finally, Google is a search engine that is widely used for job seeking in Italy, where there are no public or private employment agencies to bring the supply and demand for work together on a large scale throughout the nation and for a broad range of jobs and professions.

In any case, attention should be drawn to some limitations of GT. It does not distinguish between those seeking jobs because they have lost theirs and those who simply would like a change. It provides no data on the gender and age of the online job seekers, and multiple searches by the same individual are counted separately. In other words, GT captures only the temporal trend of the number of those seeking jobs online with no reference to their reasons for doing so or their individual characteristics.

As the monthly ISTAT labour force survey indicates a marked increase, especially among the young, in the percentage of unemployed

persons seeking jobs online (see Section 2.1), the choice of GT as a source appears to be reasonable, not least as the sole purpose of this work is to provide a forecast of the youth unemployment rate.

The forecasting models used are those typically employed in the analysis of time series: a VAR model using both official and online information and an ARIMA model estimated solely based on official data. A number of considerations regarding the choice of models are subsequently developed in Section 2.3.

2.1. Youth unemployment in Italy

The youth unemployment rate is defined at the European level as the ratio of unemployed persons age 15–24 to the corresponding labour force (employed and unemployed). It therefore represents the proportion of young unemployed people among those who are active in the labour market. The calculation of the unemployment rate thus excludes young people who are *inactive*, understood as those who are jobless but not looking or available for work, in most cases because they are still studying (inactive people account for more than 70% of the population in this age group).

The youth unemployment rate thus defined almost doubled during the recent economic crisis, rising from about 20% in 2007 to more than 40% between 2013 and 2015. Although recent months have shown a slight change in direction, the rate still is close to maximum levels, 39.1% being the latest estimate for the month of February 2016 (ISTAT, 2015a).

In addition to this marked growth, the youth unemployment rate is far higher in Italy than in the other EU countries – 42.7% in 2014 as opposed to the EU average of 22.2%. This makes the youth unemployment rate in Italy the fourth highest of the 28 member countries, preceded only by Spain, Greece, and Croatia. The position is slightly better (seventh place) with regard to the total unemployment rate: 12.7% in Italy versus the EU average of 10.2%. These few figures show that the difference between Italy and the EU is considerably greater concerning the difficulties encountered by young people in the labour market.

As Mario Draghi, president of the European Central Bank, points out in a recent interview (The Guardian, 11 March 2016), analysts also are concerned about the medium and long-term consequences of young people’s failure to participate in the labour market in terms of the loss of human capital. According to the theory of human capital, productivity on the labour market depends on various factors, among which, apart from individual abilities and the skills acquired through education and training courses, importance is also attached to what is learned on the job. As time passes, the knowledge acquired tends to become obsolete or is lost through disuse. Apart from the obvious economic and social difficulties involved, unemployment therefore also leads to the loss of human capital, affecting the future working life of the young people concerned. These elements help explain why the estimation of the youth unemployment rate is a matter of great interest at both the Italian and the European level.

The data gathered in the ISTAT labour force survey reveal considerable use of the Internet in employment seeking, especially among the young. The data related to the specific query introduced in the survey show that, with respect to the total number of people seeking work, the proportion of those using the Internet for this purpose rose from 20% in January 2004 to 60% in December 2014 (Fig. 1).

As is to be expected, given the greater use of computers and the Internet by the younger generations, these figures are higher for the sphere of youth unemployment, with a rise from approximately 25% in January 2004 to approximately 70% in December 2014. This makes it reasonable to suggest that the trend of the monthly *offerte di lavoro* (job offers) query share constitutes a useful tool for forecasting the youth unemployment rate.

It also should be noted that in December 2014, 96% of the unemployed people using the Internet examined job offers, 37% offered



Fig. 1. Unemployed people's use of the Internet in job seeking. Monthly series January 2004–December 2014. Continuous line for Internet users and dotted line for non-Internet users.

their services or replied to offers, and 12% accessed the websites of public employment offices (ISTAT, 2016). This confirms the validity of the use of *offerte di lavoro* as a keyword in the extraction of the GT series for use in forecasts.

2.2. Data description

The monthly time series of the youth unemployment rate in Italy (YUR) is considered here for the period January 2004–June 2015. The YUR series is produced based on data from the ISTAT labour force survey, the primary source of information on the Italian labour market.

The households interviewed are drawn from municipal registries by means of a sampling strategy designed to construct a statistically representative sample of the resident population in Italy with respect to the variables examined. More than 250,000 households (approximately 600,000 individuals) residents in about 1100 Italian municipalities are interviewed annually. The official estimates of the major aggregates of the labour supply are produced monthly and made public generally in the next month.

Fig. 2 plots the evolution of the youth unemployment rate in Italy for the period of analysis. The effect of the economic crisis on the participation of young people in the labour market is clearly visible. In the middle of 2008, the youth unemployment rate showed a sharp increase that continued until spring 2014. It was about 20% at the beginning of 2008, when it started to rise, and reached peaks of more than 40% in 2014. The marked variability of the series must be stressed.

The GT series – downloadable from Google by selecting a country of reference, a period of time, and one or more keywords – provides a measure of the weekly number of job searches carried out through Google. In general, for reasons connected with privacy, the indexes available online from Google Trends for any keyword are constructed as follows:

- Google Trends calculates the ratio of the number of online searches made (through the Google search engine) for a specific keyword in a given geographical area(s) on a particular day t ($K_{s,t}$) to the total number of online searches made through Google on the same day in the same area ($T_{s,t}$):

$$R_{s,t} = \frac{K_{s,t}}{T_{s,t}}$$

- This ratio then is used to calculate the weekly rate as the average of the rates $R_{s,t}$ registered on every day of a specific week γ :

$$R_{s,\gamma} = \frac{1}{7} \sum_{t=\text{Sunday}_\gamma}^{\text{Saturday}_\gamma} R_{s,t}$$

- Finally, the index $R_{s,\gamma}$ is scaled to 100 as the ratio to its value in the week in which it reaches the maximum level. The Google index (GI) for a specific keyword K , in week γ , and in area s therefore is given by

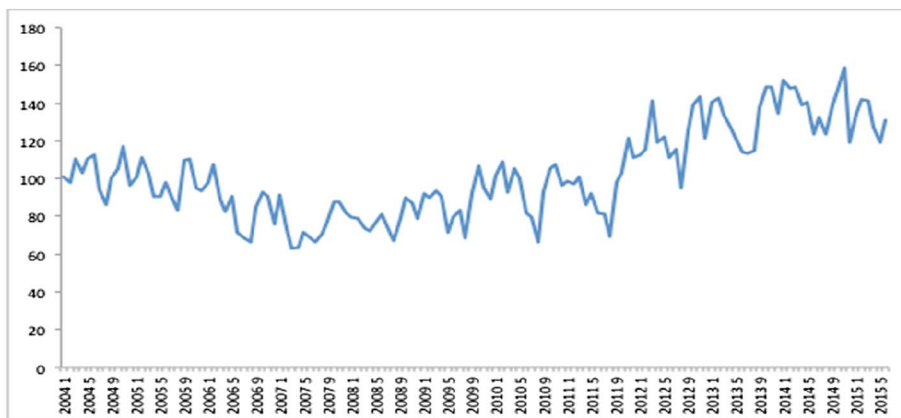
$$GI_{K,s,\gamma} = \frac{100}{\max_\gamma (R_{s,\gamma})} R_{s,\gamma},$$

which represents the percentage of Internet users making searches on Google for a specific keyword in a certain area and in a given week.

The GT series considered here coincides with the GI in which $K = \text{offerte di lavoro}$, $s = \text{Italia}$, and γ are the weeks from January 2004 to June 2015. The weekly GT data were aggregated into monthly averages in accordance with the LFS scheme of allocation of weeks of investigation to months.

Both the series considered, YUR and GT, were seasonally adjusted by means of a TRAMO SEATS procedure (Hillmer and Tiao, 1982) to eliminate seasonal fluctuations of the phenomena considered.

Fig. 2. Youth unemployment rate in Italy, 2004–2015.



Reference henceforth will be made solely to the seasonally adjusted series.

2.3. The model

Two types of time series models were estimated to ascertain if Google Trends data can be of use in forecasting the youth unemployment rate. One is an ARIMA model, using only the information of the official series for the youth unemployment rate, while the other – a VAR model using cross-correlation of the two series (Johansen, 1995) – combines the official series for the youth unemployment rate with the series of the job offers query share. It should be pointed out that in neither case is this paper concerned with putting forward a model to identify the determinants of the variability of the youth unemployment rate in Italy. It also should be noted that the ARIMA model, estimated solely on the official ISTAT series, is presented only as a benchmark. The sole purpose of the estimation of a very simple time series model is to show if, with respect to the use of the official series alone, the introduction of Internet search data can prove useful with a view to forecasting.

The use of the ARIMAX model (Hamilton, 1994) with GT as an exogenous variable is widely advocated in the recent literature on the use of big data to forecast economic indicators (D'Amuri and Marcucci, 2017; Vicente et al., 2015). In our view, however, this type of model is not appropriate in this case for at least three reasons. First, regarding the GT variable as exogenous appears to be questionable. Both the GT and YUR series are expressions of the same phenomenon, and the latter cannot be regarded as causing the former. Second, the temporal dynamics of the auxiliary series (GT) is not taken into account in an ARIMAX model. It instead is our opinion that the characteristics with which the GT series evolves over time must be considered explicitly. Its temporal dynamics is unquestionably connected with that of the youth unemployment rate to be forecasted. This does not happen in a model of the ARIMAX type, in which the GT variable, whether contemporary or delayed, is regarded as deterministic. The third reason is that ARIMAX models are not suitable for forecasting more than one step ahead unless introducing an arbitrary ad hoc delay for the GT variable linked to the forecast horizon.

The VAR model presented here, other than considering explicitly GT and YUR variables as endogenous, takes into account the temporal dynamics of both series as well as their reciprocal effects. Contrary to what happens with the ARIMAX model, the VAR model makes the simultaneous forecasting of both series possible more than one step ahead.

For this purpose an out-of-sample procedure (Tsay, 2010, pp. 215–216) was performed by setting horizons of 1 month and 3 months ($h = 1, h = 3$); as a result, one and three observations available were excluded from the estimation of the model.

The choice of the time horizons is related to the two aims pursued: to obtain preliminary monthly estimates capable of giving an indication of the expected variations in the phenomenon of interest from one month to the next prior to the completion of the official sample survey and to obtain reliable forecasts of the youth unemployment rate that can be used in the quarterly forecast models of the Italian economic system (ISTAT, 2015b). It should be noted in this connection that the high degree of intra-annual variability of unemployment in general and therefore also of Italian youth unemployment makes the use of broader time horizons inadvisable.

The ARIMA and VAR models were estimated for each of the series linked to the horizon considered and then compared by calculating the difference between the estimated values and those actually observed to obtain the mean absolute error (MAE):

$$MAE = \frac{\sum_{i=1}^{T-h} |\hat{y}_i - y_i|}{T - h} \times 100$$

where $\hat{y}_i$ is the estimate of the youth unemployment rate in month i and y_i the observation carried out by means of the official survey for the same month.

Finally, to compare the forecast ability of the estimated models, a rolling regression (D'Amuri and Marcucci, 2017) was performed. For the two forecast horizons, the distribution of the absolute error (AE) $|\hat{y}_i - y_i|$ of the two models is presented, together with the median and the Theil index (Theil, 1966) U , defined as

$$U = \sqrt{\frac{\sum_i \left(\frac{\hat{y}_i - y_{i-1}}{y_{i-1}} - \frac{y_i - y_{i-1}}{y_{i-1}} \right)^2}{\sum_i \left(\frac{y_i - y_{i-1}}{y_{i-1}} \right)^2}}$$

where $\hat{y}$ is now the forecasted value.

3. Results and discussion

3.1. Models' estimation

It is necessary when working with models for time series to evaluate the nature of the series before estimating the model itself: for each of the series the stationarity hypothesis (Hamilton, 1994, p. 45) is verified and, in case the series want to be studied together, it is necessary to verify if they present common stochastic trends.

The Johansen test (Johansen, 1991) that allows both these evaluations to be carried out was calculated on the YUR and GT series indicated, respectively, with y_{1t} and y_{2t} .

This is a sequential test: it occurs initially if the series are only integrated (null hypothesis) or integrated and linked by a co-integration relationship (alternative hypothesis). In the second test, which is carried out only if the null hypothesis of the previous one is rejected, we compare the hypothesis that they are co-integrated against the hypothesis that they are stationary.

Since the value of the test statistic (JS) for the first problem is smaller than the 95th percentile of Johansen's distribution¹ (Table 1), we conclude that the series are integrated of order one (null hypothesis) and therefore do not proceed with the second test. Since the series are integrated they are non-stationary; to solve the problem of non-stationarity, we use the first differences of the series instead of the original ones. Finally, since the series are not co-integrated, the VAR model is the most suitable to represent the joint variability of the two series.

The results of Johansen's test about the significance of the two series' joint variability also enable us to conclude that their relationship cannot be supposed spurious, and therefore we can reasonably consider the choice of the keywords *offerte di lavoro* to be appropriate.

Let Δy_{1t} and Δy_{2t} indicate the differentiated series of the original YUR and GT series ($\Delta y_{it} = y_{it} - y_{it-1}$, $i = 1, 2$). The orders of the autoregressive (AR) and moving average (MA) components for the ARIMA model, as well as the AR order for the VAR model, were chosen based on the Akaike Information Criterion (AIC) (Akaike, 1974). The ARIMA (0,1,1) model estimated on the sole basis of the YUR series therefore is:

$$\Delta y_{1t} = c_1 \varepsilon_{t-1} + \varepsilon_t \quad (1)$$

where c_1 is the unknown parameter of the model.

It has to be noted that, due to the marked variability of the series, the error component ε_t was estimated by means of a generalized autoregressive conditionally heteroskedastic process (GARCH) of order (1,1) model (Engle, 1982). The introduction of the GARCH component is due to the fact that the differentiated series of YUR and GT show the characteristics of a volatility cluster (i.e. volatility is high for certain periods and low for others) and volatility variation within fixed range

¹ The statistical test does not have a theoretical distribution but was tabulated by Johansen (1995).

Table 1
Johansen test for the YUR and GT series.

Series length	Johansen statistic (JS)	95th percentile
T-1 (Jan. 2004–May 2015)	16.31	19.96
T-3 (Jan. 2004–Mar. 2015)	16.08	19.96

Table 2
Estimate of the ARIMA model parameters.

Forecast horizon	Parameter	Estimate	Std error	t-Value	p-Value
$h = 1$	c_1	−0.5442	0.0785	−6.9262	0.0000
	AIC	990.7648			
	MAE	19.26			
$h = 3$	c_1	−0.5537	0.0794	−6.97	0.0000
	AIC	957.2055			
	MAE	18.8			

which are well approximated by the GARCH model (Tsay, 2010, p. 111). For this reason, ε_t in Eq. (1) is

$$\varepsilon_t = \mu + \sigma_t z_t \quad (1a)$$

with

$$z_t \sim N(0, 1) \text{ and } \sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

where $N(0, 1)$ indicates the standardised normal distribution.

Table 2 shows the estimate of the parameters in Eq. (1) as well as the value of the AIC and MAE.

Finally, the VAR (8) model estimated for the variables Δy_{1t} and Δy_{2t} is:

$$\Delta y_{1t} = a_1 \Delta y_{1t-1} + a_2 \Delta y_{1t-2} + a_5 \Delta y_{1t-5} + b_5 \Delta y_{2t-5} + \varepsilon_{1t} \quad (2)$$

$$\Delta y_{2t} = b_1 \Delta y_{2t-1} + b_8 \Delta y_{2t-8} + \varepsilon_{2t}$$

where a_i $i = 1, 2, 5$ and b_j $j = 2, 5, 8$ are the unknown parameters of the model.

In Eq. (2) the error component also was estimated in each equation on the basis of the GARCH (1,1) model (Eq. (1a)).

Because $\Delta y_{it} = y_{it} - y_{it-1}$, $i = 1, 2$, Eqs. (1) and (2) can be easily written in terms of the original variables y_{1t} e y_{2t} . So Eq. (1) becomes

$$y_{1t} = y_{1t-1} + c_1 \varepsilon_{t-1} + \varepsilon_t$$

and equations in (Eq. (2)) are

$$y_{1t} = (1 + a_1)y_{1t-1} + (a_2 - a_1)y_{1t-2} - a_2 y_{1t-3} + a_5 y_{1t-5} - a_5 y_{1t-6} + b_5 y_{2t-5} - b_5 y_{2t-6} + \varepsilon_{1t}$$

$$y_{2t} = (1 + b_1)y_{2t-1} - b_1 y_{2t-2} + b_8 y_{2t-8} - b_8 y_{2t-9} + \varepsilon_{2t}$$

This makes it easy to see that the nature of the two models is very similar. In fact, in both cases the fundamental component is autoregressive: The youth unemployment rate at a certain time t is expressed as a linear function of his past observations. This linear relationship is present both in the ARIMA and VAR models.

In the ARIMA model, we estimate only the autoregressive equation for the youth unemployment rate; in the VAR model, we study a system of two equations. The first equation is still an autoregressive equation for the youth unemployment rate to which the GT component is added, while the second is an autoregressive equation for the GT variable. We therefore can see the VAR model as an extension of the ARIMA model; this extension follows an ‘incremental’ logic that consists of moving from a simpler model to a more complex one (introducing the GT auxiliary variable) if we obtain better results. The advantage of adopting the VAR model instead of the ARIMA one is that the estimation procedure exploits the link between the two series, which should lead – as in the case proposed in this paper – to a more precise estimation of the unknown parameters of the model and therefore to a

Table 3
Estimate of the VAR model parameters.

Forecast horizon	Parameter	Estimate	Std error	t-Value	p-Value
$h = 1$	a_1	−0.4084	0.0802	−5.0906	0.0000
	a_2	−0.2877	0.0799	−3.6005	0.0005
	a_5	−0.2218	0.0762	−2.9101	0.0043
	b_5	1.0695	0.2382	4.4906	0.0000
	b_1	−0.2597	0.0834	−3.1135	0.0023
	b_8	0.1862	0.0797	2.3360	0.0211
	AIC	6.72			
	MAE	18.07			
$h = 3$	a_1	−0.4127	0.0798	−5.17	0.0000
	a_2	−0.2927	0.0797	−3.67	0.0000
	a_5	0.2580	0.0783	−3.29	0.0013
	b_5	1.0605	0.2386	4.44	0.0000
	b_1	−0.26023	0.0851	−3.058	0.0027
	b_8	0.1826	0.0808	2.2593	0.0257
	AIC	6.73			
	MAE	9.41			

more precise forecasting of the phenomenon under study.

To verify if an increase in the complexity of the model corresponds to an improvement in results, the two models were assessed based on the forecasting error measures presented in Section 2.3. In both cases, the estimated parameters were highly significant. The ARIMA model, which makes no use of Google Trends information (Table 2), presented higher values of MAE than the VAR model (Table 3).

An out-of-sample procedure was applied to evaluate the accuracy of the forecast obtained with both of the models considered. The comparison is summarized in Table 4, which presents the values actually observed and those forecast by the two models together with the corresponding value of the AE.

The gain in accuracy obtained with the VAR model is more than two percentage points for $h = 1$ (April 2015) and more than one for $h = 3$ (June 2015). This shows that the forecast obtained with the support of the GT series is more accurate – in terms of forecast error – than that obtained solely based on information from the official sample survey. A rolling regression procedure was applied, as described in Section 3.2, to assess the robustness of the results, that is, to ascertain if the greater accuracy of the forecasts obtained with the VAR model is maintained with the transition from one month to another (e.g. January and March are forecast in December, while February and April are forecast in January, and so on following the order of the months).

3.2. Rolling estimation and forecast procedure

Official statisticians frequently are confronted with the routine problem of forecasting socio-economic indicators such as the youth unemployment rate. The procedure generally in use is the one of estimating a time series model based on the observed values up to time t and then producing the desired forecast one or more steps ahead. The same happens for the following time, when the model is newly estimated on the observed values up to $t + 1$ and the forecast produced. A rolling regression was carried out to simulate this procedure to evaluate the forecasting ability of the two models.

To obtain a global evaluation of the VAR and ARIMA models' performance, the AEs were computed for each of the forecasted values on the basis of the rolling regression. Because the two distributions proved to be strongly asymmetric (Fig. 3a and b), the median was considered to

Table 4
Actual and forecast values of the youth unemployment rate.

Month	Actual value	ARIMA value	ARIMA AE	VAR value	VAR AE
April 2015	127.3	133.2	5.9	124.0	3.3
June 2015	131.2	120.6	10.6	122.8	9.4

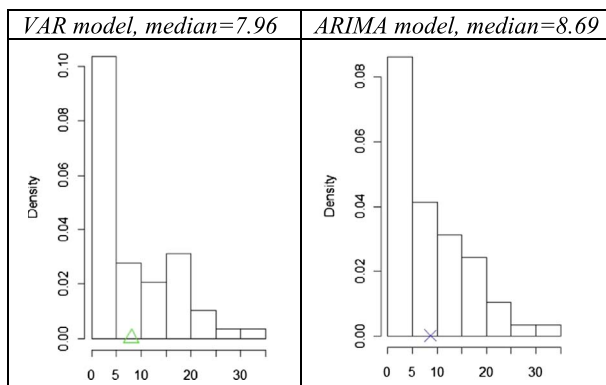
Fig. 3. Absolute error distribution, $h = 1$.

Table 5

Theil index for the VAR and ARIMA models.

Forecast horizon	VAR model	ARIMA model
$h = 1$	0.775	0.789
$h = 3$	1.067	1.106

make a comparison between them.

To the same extent, the Theil index was computed and is reported in Table 5.

In the case of $h = 1$, the procedure consists of obtaining the forecast of the series of interest month by month and inserting the value actually observed each time before proceeding to the next step of estimation and forecasting.

The first 80 observations (from January 2004 to October 2010) of the YUR and GT series initially were considered to estimate the two models. A forecast for November 2010 then was obtained by means of the fitted models. This procedure was repeated, inserting the real value month by month and obtaining the following month's forecast for each of the months from November 2010 to June 2015.

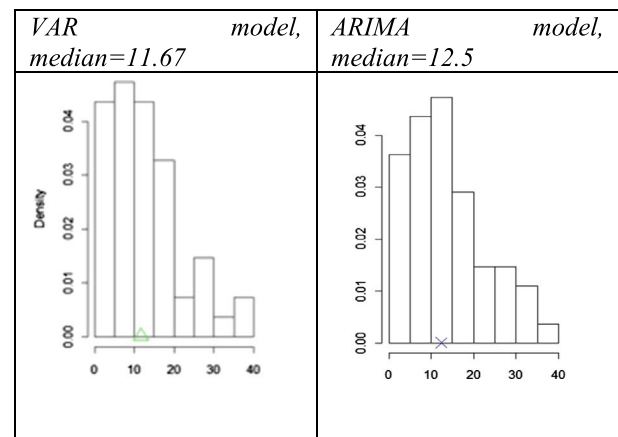
The median value of the AE distribution for the VAR model ($h = 1$) is lower than the corresponding one for the ARIMA model (Fig. 3). This means that the AE of the VAR model assumes smaller values more frequently and that this model provides a more accurate estimate of the youth unemployment rate more frequently than the ARIMA one. The better performance is confirmed by the Theil index (Table 5), which emerges as lower for the VAR model than for the ARIMA model.

For the VAR model, when $h = 3$ (i.e. when the forecast is three steps ahead), both variables must be forecasted for $t + 1$ and $t + 2$ to obtain the forecast for $t + 3$.

Even in this case, the results obtained are better for the VAR model than for the ARIMA one (Fig. 4), although the reduction in the median of the absolute error from the ARIMA to the VAR model is lower than before. The Theil index also confirms the better performance of the VAR model.

4. Conclusions

In Italy, as in the other European countries, analysts of the labour market pay great attention to studying the trend of unemployment in general and youth unemployment in particular, above all from a short-term perspective. It therefore is important to provide indications of the trends in these aggregates that are as accurate and reliable as possible to permit correct analysis of the national situation and use of the data for effective policymaking by the relevant authorities. Hence, the purpose of this paper was to ascertain the possibility of improving the forecasts of youth unemployment in Italy by combining the official sources with big data.

Fig. 4. Absolute error distribution, $h = 3$.

Such forecasts are of use to the ISTAT for two reasons. First, they constitute advance estimates of the phenomenon under examination and can be used for preliminary analysis before the official estimates have been produced. Second, they can be used within the framework of macro-econometric forecasting models to provide indications of the prospects for the economic system as a whole.

The results confirm the close connection between the level of youth unemployment in Italy and the use of the Internet to look for work, so this method can be used as auxiliary information in forecasting, at least for short and medium forecast horizons.

The VAR model, characterized by the use of the GT series, produces better results in terms of forecast error than the ARIMA, which is based exclusively on information from the official series for the youth unemployment rate. This suggests that a snapshot of what is happening in society, although not accurate in terms of statistical quality, provides readily available, low-cost information that can be used to draw at least some preliminary conclusions about an indicator of great social, political, and economic importance. This confirms the initial idea that what can be exploited in terms of forecasting capacity of the GT series is the way in which it moves in unison with the youth unemployment rate series and thus reflects the changes in the population manifested through its use of the Internet.

In the light of these considerations, it appears to be possible to conclude that the use of GT data as an auxiliary variable – together with a suitable forecasting model – is a good way to obtain reliable forecasts of the unemployment rate. At the same time, however, the fact cannot be overlooked that the level of forecast error is by no means negligible. In other words, although the results obtained through the use of GT data are encouraging and suggest that we should continue in this direction, the procedures for extracting information from GT and the possible application in other areas of estimation require further investigation.

It should be pointed out that, given the nature of the GT and the YUR data, the procedure put forward here is replicable for small-scale contexts, such as particular categories of labour and/or specific territorial and temporal spheres. Also, the model proposed here was applied to the unemployment rate series in Italy, which presents situations of marked variability, making it difficult to isolate the error component effectively from the structural one due to the economic crisis that has struck Italy harder than other European countries in the last few years.

Finally, a general remark seems to us to be necessary. The use of Google Trends in official statistics as an aid to statistical production, or for the production of additional information, is a great opportunity and a challenge for the national statistical institutes, but it implies the need for a careful control of the quality of information. Although the debate often is focused on the issues of information volume, the velocity at which it can be acquired, and its variety, national statistical offices also

must ensure that estimates are accurate. For this reason, as statistical authorities, they are responsible for drawing up guidelines and methodologies, and promoting the exchange of knowledge between statistical offices, academics and owners of private data sources (European Commission, 2013).

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